

Evidence-Led Sensor Architecture for a Multimodal Prosthetic-Liner Slab

Abstract

This review translates prior work on prosthetic-interface monitoring and printed sensors into design requirements for a 56 mm × 55 mm × 5.2 mm silicone test slab. Rather than treating each source as an isolated summary, the literature is evaluated against four design questions: what must be measured, which geometry is transferable, what performance has already been demonstrated, and what must be revalidated after encapsulation. The resulting architecture combines one four-capacitor normal/shear sensor, two embedded TMP36 devices, four printed resistive temperature sensors, and four printed capacitive humidity sensors.

1 Interface problem and design objective

A prosthetic liner is both a mechanical interface and a thermal barrier. Normal pressure supports load transfer, while relative movement generates shear. At the same time, the socket-liner system restricts heat and moisture transport. Pressure alone therefore cannot describe the conditions associated with discomfort, slipping, perspiration, or skin risk [1, 2, 7].

The immediate objective is not yet a complete wearable liner. It is a controlled slab demonstrator that can answer whether multiple sensing principles can coexist inside silicone without losing sensitivity, becoming cross-sensitive, or producing unacceptable local thickness. The literature is used here to select candidate geometries and realistic starting dimensions; all quoted performance remains source performance until reproduced on the fabricated slab.

2 Evidence-to-design matrix

Table 1 states the engineering contribution of each principal source and, importantly, what remains unproven for the present slab. This separates transferable evidence from assumptions that require new testing.

3 Mechanical sensing architecture

3.1 Four-capacitor normal/shear cell

The strongest transferable element from Albrecht *et al.* is the differential geometry, not the published sensitivity in isolation [3]. Four overlapping electrode regions form C_1 – C_4 . Compression changes the plate separation and produces a common-mode capacitance change. Translation along x or y increases overlap in one opposing capacitor while decreasing

it in the other. Appropriate sums and differences can therefore estimate normal loading and two shear components.

The reported electrode features—approximately 210 μm fingers, 40 μm spacing, and a 40 μm dielectric—are defensible starting values. They are not automatically optimal for the present silicone system. Encapsulation changes effective stiffness and load transfer; cable capacitance can also be large relative to a femtofarad-scale response. The first mechanical experiments must therefore report a calibration matrix rather than a single sensitivity:

$$\begin{bmatrix} \Delta C_1 \\ \Delta C_2 \\ \Delta C_3 \\ \Delta C_4 \end{bmatrix} = \mathbf{K} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} + \epsilon.$$

The off-diagonal terms of $\mathbf{K}$ quantify cross-axis coupling, while ϵ includes drift and measurement noise.

4 Thermal and humidity architecture

4.1 Printed resistive temperature sensor

Liew and Lee demonstrate a printable serpentine resistor compatible with a PET carrier and Dimatix deposition [5]. The long conductive path raises resistance within a small footprint. The slab adopts the geometry and material logic, but its calibration window should focus on prosthetic-interface temperatures rather than the source study's full 30 °C to 100 °C range.

4.2 Printed interdigitated humidity sensor

The humidity design uses two interleaved electrode combs coated with CAB [4]. Water uptake changes the effective dielectric environment and thus capacitance. Unlike the temperature trace, this device cannot be fully sealed in silicone: the sensing film requires vapour access. A porous window or selective protective layer is therefore a design requirement, not an optional detail.

4.3 Embedded TMP36 references

The TMP36 provides a simple voltage output of 10 mV/°C, operates from 2.7 V to 5.5 V, and is specified to approximately ± 2 °C over its full temperature range [8]. Two devices give independent reference points for comparison with the printed traces. Their TO-92 packages are nevertheless approximately 2.8 mm thick inside a 3.5 mm carrier layer. Local stiffness, coverage above and below the package, and thermal lag must therefore be measured rather than assumed acceptable.

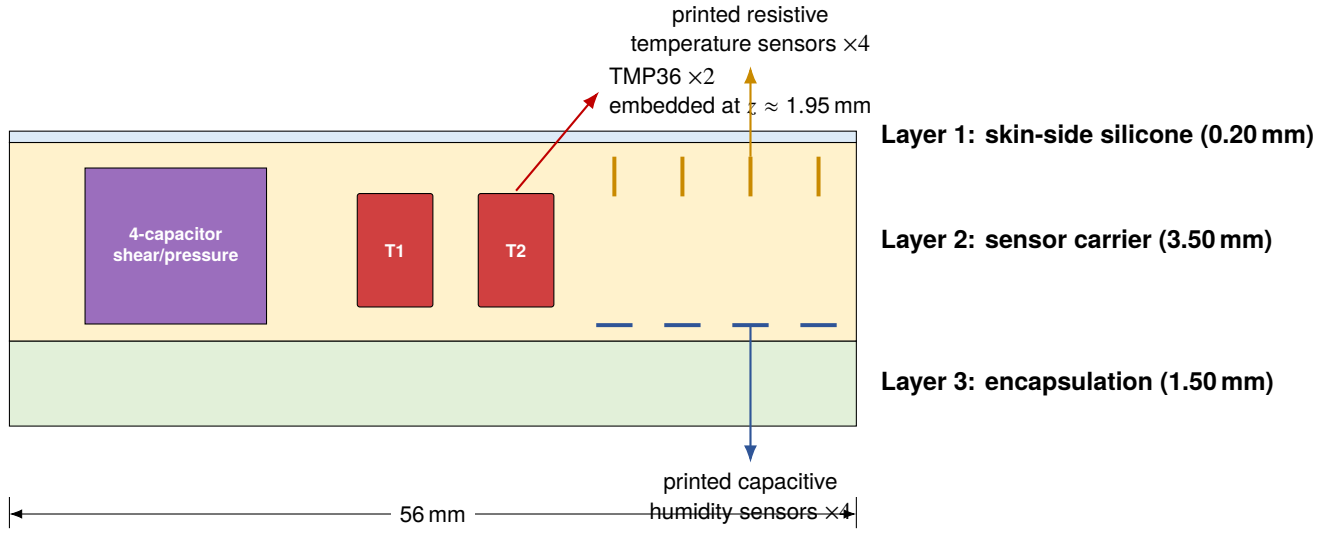


Figure 1: Proposed slab cross-section and sensor allocation. Vertical scale is exaggerated for clarity; sensor positions are schematic rather than manufacturing tolerances.

Table 1: How the literature informs the demonstrator.

Source	Measured quantity	Key demonstrated result	Design decision adopted here	Must be revalidated
Albrecht <i>et al.</i> [3]	Normal force and biaxial shear	Four differential capacitors separated normal and directional shear; ~ 5.2 fF/N normal and ~ 13.1 fF/N shear sensitivity over 0.1 N to 8 N.	E-shaped overlapping electrodes, four-capacitor arrangement, printed conductors, thin elastomeric dielectric.	Sensitivity after silicone encapsulation, axis decoupling, parasitic capacitance, hysteresis, and fatigue.
Ait-Mammar <i>et al.</i> [4]	Relative humidity	CAB-coated interdigitated capacitor responded over approximately 20–90% RH.	Four distributed interdigitated sensors; nominal 100 μ m gaps and CAB-sensitive film.	Moisture access through the liner, liquid-water protection, drift, condensation response, and encapsulation effects.
Liew and Lee [5]	Temperature	Printed serpentine resistors on PET showed an approximately linear resistance–temperature response; the relevant ink produced $\sim 0.0543 \Omega/^{\circ}\text{C}$.	Four PET-supported serpentine traces with 0.5 mm tracks and 0.4 mm gaps.	Calibration near skin temperature, strain cross-sensitivity, lead resistance, thermal lag, and sensor-to-sensor repeatability.
Ghoseiri <i>et al.</i> [6]	Liner temperature field	Twelve sensing positions showed the value of spatial and differential temperature monitoring in a prosthetic system.	Two embedded reference-temperature points to compare local thermal behaviour.	Local stiffness, heat conduction through silicone, and whether two points give adequate spatial information.
Paternò <i>et al.</i> [7]	Temperature and humidity in use	Twelve integrated Hygrochron devices captured spatial changes during a prosthetic-liner trial.	Integrated, multi-location monitoring and comparison against a conventional reference condition.	Miniaturisation, multimodal integration, comfort, durability, and extension from one case study.

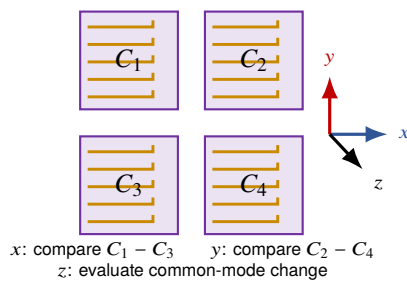


Figure 2: Functional interpretation of the four-capacitor cell, schematically adapted from Albrecht *et al.* [3]. The drawing explains signal separation and is not a fabrication mask.

Resistive temperature **Capacitive humidity cell**

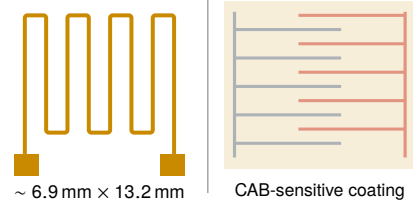


Figure 3: Source-informed printed geometries used as starting points for the slab. Left: serpentine resistive temperature sensor [5]. Right: CAB-coated interdigitated humidity sensor [4].

5 Adopted design inputs

Table 2: Current source-informed starting values.

Element	Starting specification
Slab envelope	56 mm × 55 mm × 5.2 mm
Silicone layers	0.20 mm, 3.50 mm, and 1.50 mm
Shear/pressure cell	C ₁ –C ₄ ; 210 µm fingers; 40 µm gaps; 40 µm dielectric
Printed RTD	~ 6.9 mm × 13.2 mm; 0.5 mm track; 0.4 mm gap; PET carrier
Humidity cell	52 interdigitated pairs; 100 µm gap; ~ 8 µm CAB coating
Reference sensors	TMP36 × 2, $z \approx 1.95$ mm

These values establish a reproducible first prototype; they are not final optimised parameters. Each must be tied to a fabrication tolerance and then checked after curing and encapsulation.

6 Required validation and research gap

The combined architecture addresses a gap left by the individual studies: thin, colocated measurement of mechanical loading, temperature, and humidity within one silicone structure. The decisive contribution will not be the mere presence of 11 sensing elements. It will be evidence that the elements remain interpretable after integration.

The minimum validation programme is therefore:

1. dimensional inspection of all printed features and cured layers;
2. normal/shear calibration including cross-axis terms;
3. temperature calibration before and after encapsulation;
4. humidity calibration with a controlled vapour-access condition;
5. thermal-lag and moisture-response tests;
6. cyclic loading, drift, hysteresis, and repeatability assessment; and
7. inspection for debonding, conductor cracking, or silicone tearing.

Paternò *et al.* show that distributed thermal and humidity monitoring can be incorporated into a customised liner [7]; Ghoseiri *et al.* show the value of spatial temperature measurement [6]. The present work extends that logic toward thinner printed elements and simultaneous shear/pressure measurement. Progression to a wearable liner is justified only if the slab demonstrates acceptable mechanical integrity, stable calibration, and no problematic local stiffness.

7 Conclusion

The literature supports a specific multimodal prototype rather than a generic “smart liner.” The four-capacitor cell provides a route to directional mechanical sensing; printed serpentine and interdigitated devices provide distributed temperature and humidity measurements; and two TMP36 devices provide embedded references. The main unresolved issue is integration. Silicone encapsulation, mechanical strain, moisture access, parasitic capacitance, and component thickness can all change

source-level performance. Accordingly, the slab should be presented as a validation platform whose purpose is to measure those integration effects before a curved wearable device is attempted.

References

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